

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 3

The Map of Mathematical Fields

Exercises

Self-Study Curriculum

1. Take the concept of “distance.” In geometry, distance is a specific number (Euclidean distance). In topology, distance is irrelevant, only neighborhoods matter. In analysis, distance is generalized to a *metric* (any function satisfying three axioms). In information theory, distance between probability distributions is measured by KL-divergence. Place each of these on the hierarchy of structure preservation. Which “distance” is the most general? Which is most specific? What do you lose and gain as you move along this hierarchy?

2. You use matrix decomposition regularly in data science (SVD, PCA, eigendecomposition). These decompositions can be viewed algebraically (factoring a matrix into simpler matrices), geometrically (decomposing a transformation into rotations and scalings), or analytically (finding the dominant modes of a system). For one decomposition you know well, write a paragraph from each perspective. Which perspective gives you the most insight for the kind of work you do?

3. The Euler characteristic of a shape (vertices - edges + faces) is a topological invariant. A cube and a sphere have the same Euler characteristic (2). A donut has a different one (0). Why would a topologist say “a cube is the same as a sphere”? What has been preserved and what has been thrown away?
